Cartify: A Scalable Quality-Gated Catalogue Pipeline and Attention-Based Recommendation Engine for Verified Electronic Listings

Dhiraj Pawar, Kalpesh Sawkare, Sainath Nanaware, Preet Panaviya
Department of Computer Engineering, Vidyalankar Institute of Technology, Mumbai

Abstract—Electronic waste mitigation relies heavily on circular commerce and certified secondary product remarketing. In this paper, we propose Cartify, an end-to-end catalogue engineering architecture featuring a quality-gated verification pipeline that filters 572,083 raw product entries down to 17,931 verified electronics listings. The pipeline feeds a star-schema analytical warehouse paired with a four-model attention fusion recommender. Empirical benchmarks show Cartify achieves a Hit Ratio@10 of 0.327 compared to 0.260 for standard Neural Collaborative Filtering (NCF), providing a 25.7% accuracy improvement.

I. INTRODUCTION

Circular electronics remarketing faces severe friction due to metadata degradation, duplicated product listings, and mismatched buyer intent. Standard collaborative filtering architectures degrade significantly when processing sparse secondhand inventories. Cartify resolves this with a dual-stage architecture: rigorous data deduplication and attention-weighted neural fusion.

II. EXPERIMENTAL RESULTS

Model Architecture	Hit Ratio@5	Hit Ratio@10	NDCG@10	Latency (ms)
ItemKNN Baseline	0.142	0.215	0.118	4.2 ms
Matrix Factorization (BPR)	0.165	0.238	0.134	8.1 ms
Neural Collab Filtering (NCF)	0.188	0.260	0.152	16.4 ms
Cartify Attention Fusion (Ours)	0.245	0.327	0.198	12.8 ms